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Electronic Practical Work

Microcontrollers Security

Reading Protection

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Programs are increasingly intricate and several teams or companies can collaborate in an embedded system project. In the case of companies that want to protect their IP, a company A develops functions used by a company B, but the company B must not be able to read the program, otherwise the IPs of the company B are no longer protected. Hence, a mechanism is mandatory to execute a section of the flash memory while it is impossible to read this given section of the flash. With the code that runs in the FLASH, it is therefore necessary to segment this memory and to achieve this result, the following skills must be acquired:

- PCROP concept
- Configure the link file (linker Script)
- Option Byte configuration via the bootloader

In order to apply these concepts, the developed IP that must be protected is the change of status of the diode.

Work to be provided

- ✓ Create a new project called pcrop with the nucleo-L476
- ✓ In this project, create a new source file called 'led_blinking.c' and write the program below

```
#include "main.h"

#define LED_PORT_GPIO_ADDRESS *((uint32_t *) 0x48000014)
#define GPIO_PIN_LED    1<<5


void secret_led_blinking(void) {

    LED_PORT_GPIO_ADDRESS ^= GPIO_PIN_LED;
}
```

- ✓ In main.c, call secret_led_blinking

```
/* USER CODE BEGIN WHILE */
while (1)
{
  /* USER CODE END WHILE */

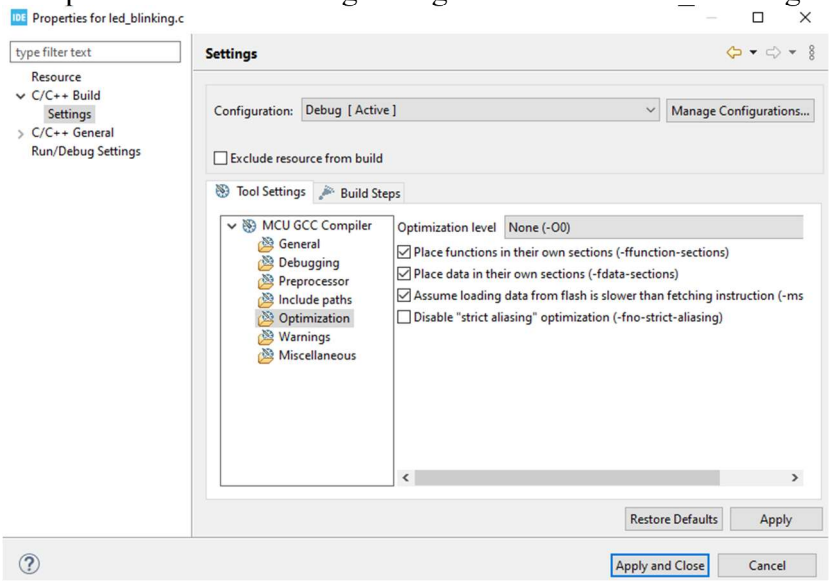
  /* USER CODE BEGIN 3 */
    secret_led_blinking();
    HAL_Delay(500);
}
/* USER CODE END 3 */
```

- ✓
- ✓ Run the program and retrieve the assembly code. Check the location of the secret_led_blinking function.
- ✓ The goal is to hide the code. To do this, it is necessary to put the code in a section of the FLASH that will be protected. In the STM32L476RGTX_FLASH.ld file
  - In the linker file, create a new memory area called PCROP. This area is only executable, starts at 0x0800 8000 and is 16K long.
  - In the Flash area, set the size to 16K as well
  - Put the code below in the STM32L476RGTX_FLASH.ld file to force the led_blinking program to be executed

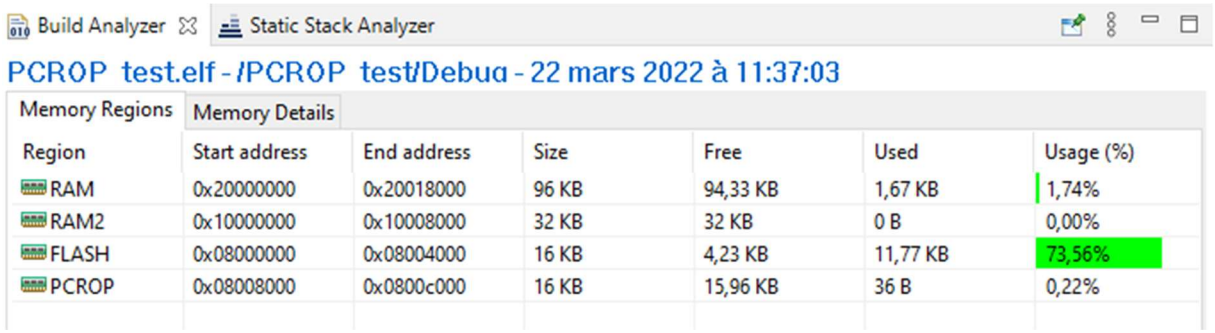
```
/* The program code and other data into "FLASH" Rom type memory */
.PCROPed :
{
  . = ALIGN(4);
  *led_blinking.o (.text .text*)
  . = ALIGN(4);

} > PCROP
```

- ✓ Compile with the following configuration for the led_blinking file

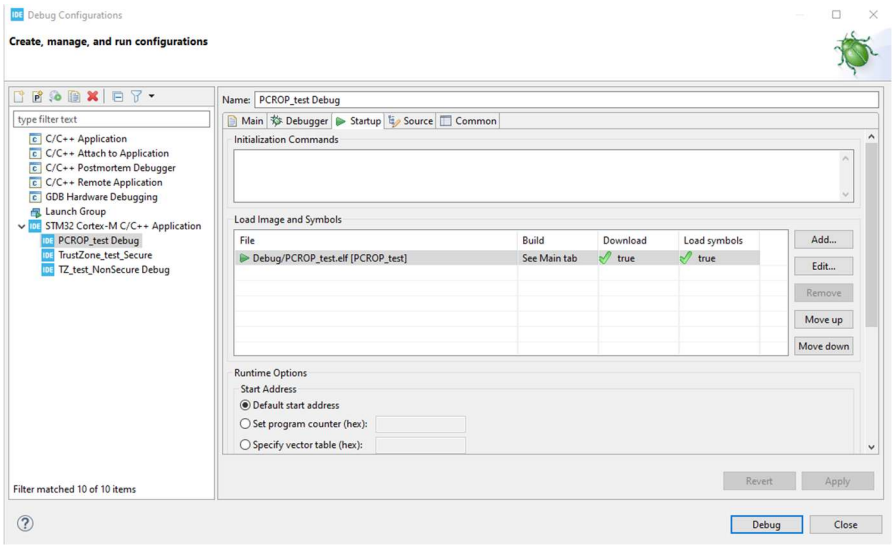


- ✓ Restart the program and check that the secret_led_blinking function has changed location. In the build analyzer, it is possible to view the different memory areas and the used size.

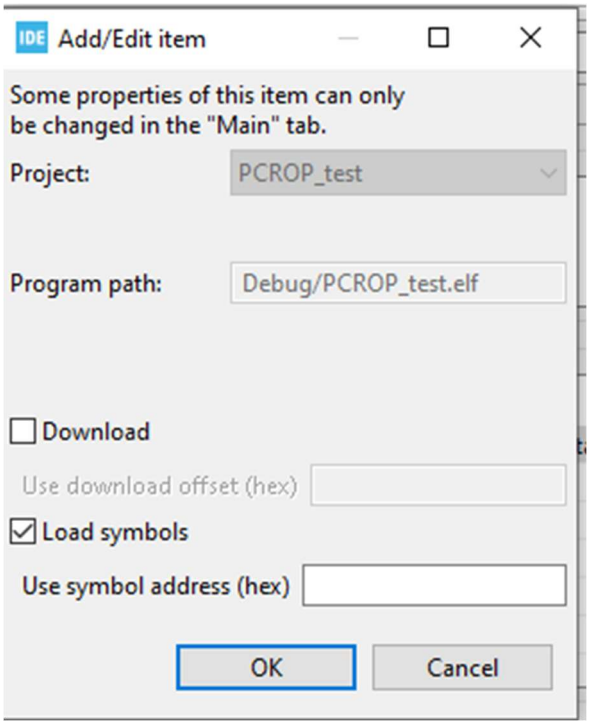


Region	Start address	End address	Size	Free	Used	Usage (%)
RAM	0x20000000	0x20018000	96 KB	94,33 KB	1,67 KB	1,74%
RAM2	0x10000000	0x10008000	32 KB	32 KB	0 B	0,00%
FLASH	0x08000000	0x08004000	16 KB	4,23 KB	11,77 KB	73,56%
PCROP	0x08008000	0x0800c000	16 KB	15,96 KB	36 B	0,22%

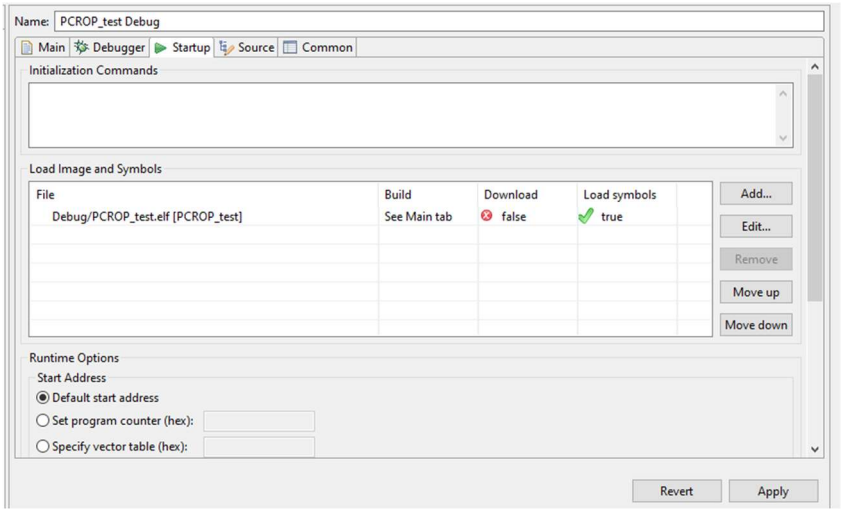
- ✓ When the code works perfectly, the last step is to hide the code, or in other words, prevent it from being read.
- ✓ Read the operation of PCROP in section 3.5.2 of the reference manual in order to understand the registers to program and determine their values to protect the PCROP area create the link file (remember that the Flash is segmented by 2KB pages)
- ✓ Program them using the cubeProgrammer
- ✓ When PCROP is activated, the program no longer works. Press the reset button and check that the LED flashes again.
- ✓ In order to test the program in the CubeIDE, you must avoid erasing the memory and re-downloading the program into the microcontroller as this is impossible with PCROP activated.
    - Go to the debug configuration and the Startup tab



- Edit and remove download selection



- The window should pop up as below



- ✓ Restart the debug and check the assembly code of the secret_led_blinking function. What to notice?
- ✓ Cancel the PCROP by programming the RDP with the cubeProgrammer.